

Exponential Modeling

SUPPORTING SKILLS LEGEND

Student learning objectives for this unit

I — Introduce

D — Deepen

A — Apply

R — Reinforce

- ☐ 1. I can **distinguish** linear and exponential change using rates, multipliers, tables, graphs, or contexts.

RELATED SKILLS

- D Compare functions shown in different representations.
- D Use constant differences or ratios to distinguish linear and exponential patterns.

- ☐ 2. I can **construct** an exponential function from two data points, a table, or a context.

RELATED SKILLS

- D Determine an exponential model from tabular values.
- D Identify initial amount, rate, and time interval in context.
- D Solve for an exponential model using two points.

- ☐ 3. I can **interpret** the initial value, growth or decay factor, and growth or decay rate of an exponential model.

RELATED SKILLS

- D Convert a percent increase or decrease to a growth or decay factor.
- D Explain a model parameter in the language of its context.
- R Recall and correctly use exponential-function terms including initial value, growth or decay factor, growth or decay rate, and asymptote.

- ☐ 4. I can **determine and interpret** doubling time or half-life from an exponential model.

RELATED SKILLS

- D Translate a doubling-time or half-life question into an exponential equation.
- A Identify initial amount, rate, and time interval in context.

- ☐ 5. I can **solve** exponential and logarithmic equations in modeling contexts.

RELATED SKILLS

- A Reject solutions that make a logarithm argument nonpositive.
- R Combine or isolate logarithms before converting to exponential form.
- R Isolate an exponential expression and apply a logarithm.

- ☐ 6. I can **fit** an exponential regression model using technology and evaluate its fit.

RELATED SKILLS

- I Use technology to determine an exponential regression model.
- D Use residual patterns and contextual evidence to judge whether a model is appropriate.
- A Interpret the sign and magnitude of a correlation coefficient in context.

☐ **7. I can **compare** linear and exponential models and justify which is more appropriate for a data set or context.**

RELATED SKILLS

- I** Select a function family whose behavior fits a table, graph, or context.
 - D** Use residual patterns and contextual evidence to judge whether a model is appropriate.
 - A** Compare functions shown in different representations.
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